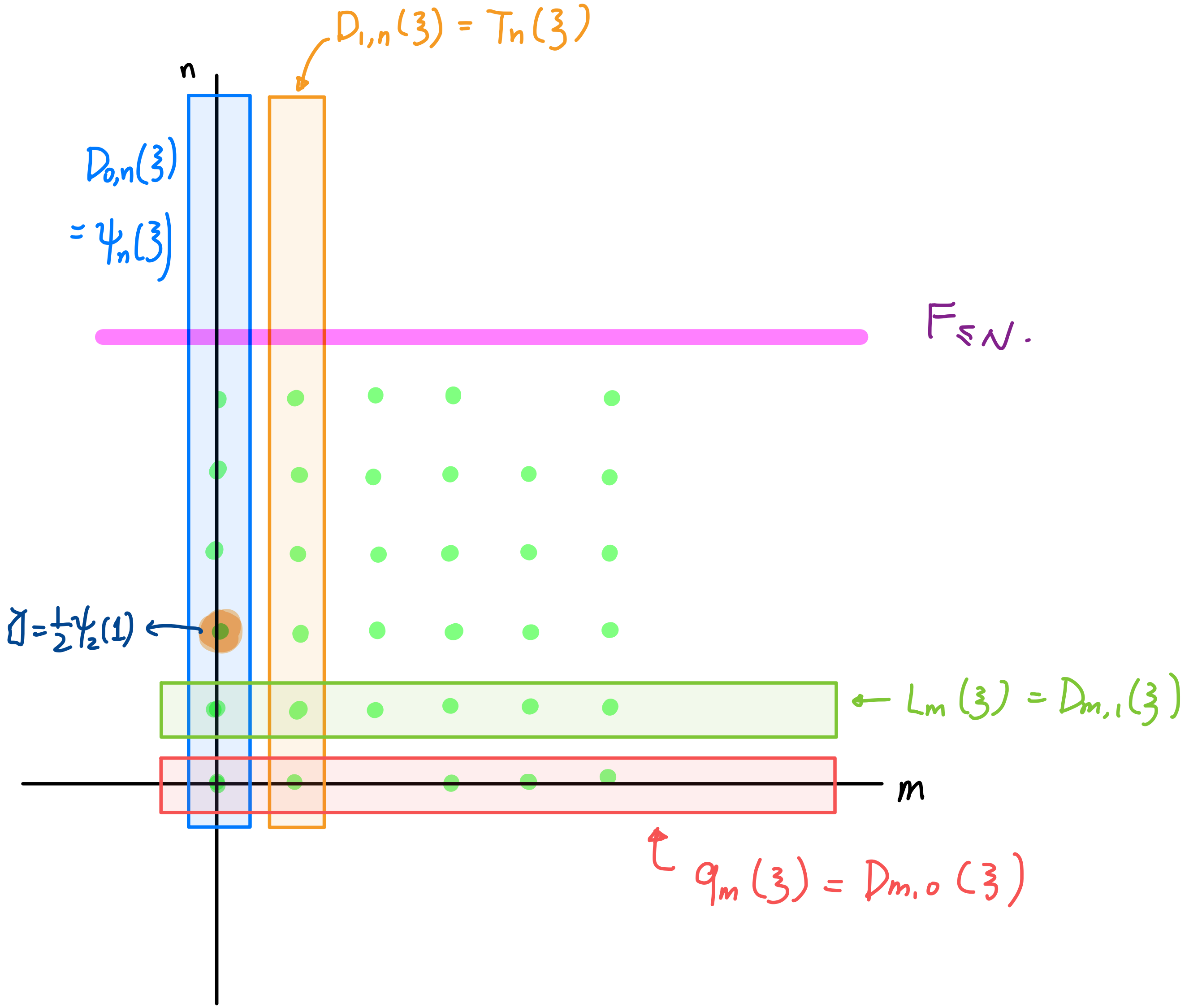
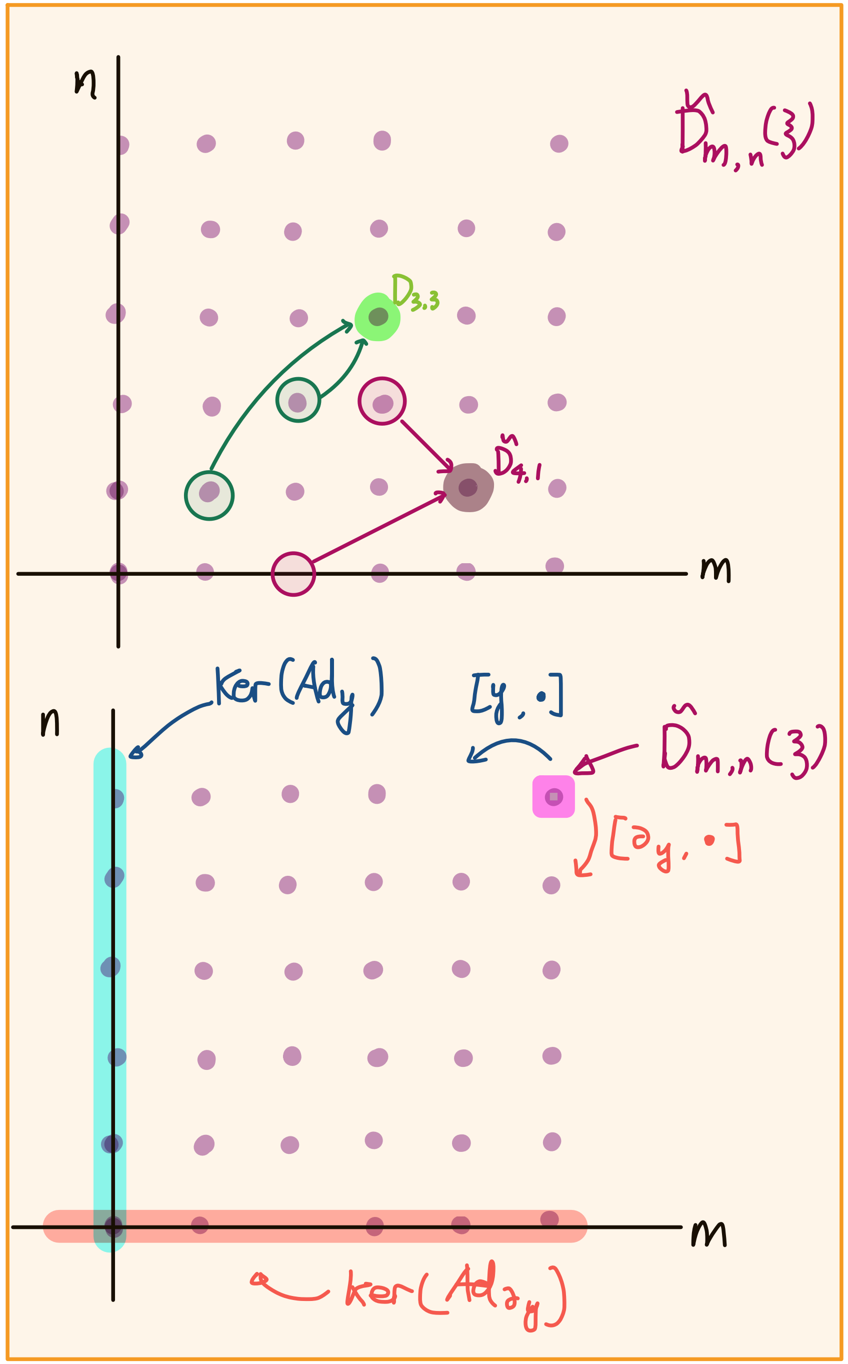

$D_{1,n}(\xi) = T_n(\xi)$
n
$D_{0,n}(\xi)$
$= \psi_n(\xi)$
$F \leq N.$
$\gamma = \frac{1}{2}\psi_2(1)$
$L_m(\xi) = D_{m,1}(\xi)$
m
$q_m(\xi) = D_{m,0}(\xi)$


n
$\hat{D}_{m,n}(\xi)$
$D_{3,3}$
$\hat{D}_{4,1}$
m
$\mathrm{Ker}(Ad_y)$
$[y, \bullet]$
n
$\hat{D}_{m,n}(\xi)$
$[\partial_y, \bullet]$
m
$\mathrm{ker}(Ad_{\partial_y})$

$$
\begin{tikzcd}
 & \mathcal{M}^{ell}_{r,d,D} \arrow[r, hook] \arrow[loop above, "W^{ell}/H"] & \mathcal{M}_{r,d,D} & & & & & & \\
\mathcal{M}^{parell}_{r,d-1,D} & \mathcal{M}^{parell}_{r,d,D} \arrow[l, "\mathrm{Mod}"'] \arrow[u, "\mathrm{fgt}"] \arrow[r, hook] \arrow[d, "\text{“reduced”}"'] \arrow[loop above, "W^{parell}/H_{r,D}"] & \mathcal{M}^{par,K}_{r,d,D} \arrow[d] & & \overset{(\text{projectivizat}^{\circ})}{\overline{\mathcal{M}^{par}_{r,d,D}}} & & \chi_D^{-1}(0) \arrow[ll] \arrow[d, equal] & \widetilde{\mathcal{M}}_{r,d} \arrow[dl, "\iota"'] \arrow[dr, "\pi"] & \\
W_{red} \quad G & \mathcal{M}^{parell}_{r,d,D} \arrow[r, hook, "\iota^{parell}"] \arrow[d, "\chi"'] & \mathcal{M}^{par,K}_{r,d,D} \arrow[urr] \arrow[d, "\bar{\chi}"] \arrow[drrr, "\chi_D"] & & & \chi_D^{-1}(\mu) \arrow[lll, "\iota_\mu"'] \arrow[r, squiggly] \arrow[d, "\chi_D"] & \mathcal{M}^{par,nilp}_{r,d,D} \arrow[rr, "(D=(p))"'] \arrow[d] & & \mathcal{M}_{r,d} \\
 & \mathbb{A}^{parell} \arrow[r, hook, "\text{open}"] & \overline{\mathbb{A}}^{par} \cong \mathbb{A} \times \mathbb{A}_D \arrow[r, "\mathrm{pr}"] & \mathbb{A}_D & & \mathbb{A}^{r|D|-1} \ni \mu \arrow[ll] \arrow[r, squiggly, "(a)\ \mu \mapsto 0"'] & 0 & & \\
 & & & \mathrm{Maps}(D, \mathfrak{t}) \arrow[u, equal] & & \mathrm{Maps}(D, \mathfrak{t}) \arrow[u, equal] & & &
\end{tikzcd}
$$

§ Surface W-algebra asso. to. orbicurve. case when $r=2$

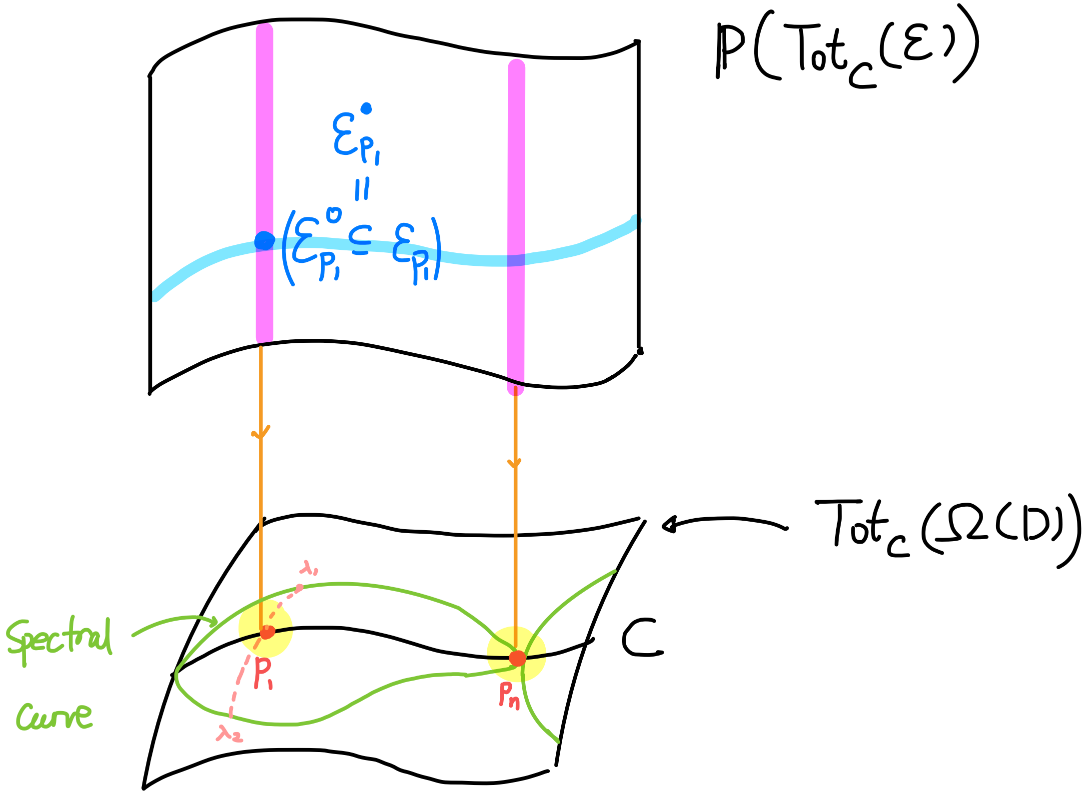
$$\mathcal{M}^{par}_{r,d,D} \ni (\mathcal{E}, \theta, (\mathcal{E}^\bullet_{p_i})_{i=1,\dots,n})$$

lying over

$$\boxed{C_{(r,D)}} := C \times_{[\mathbb{A}^1/\mathbb{G}_m],\theta_r} [\mathbb{A}^1/\mathbb{G}_m]$$

orbicurve

w/. "folding" at supp (D)



For the case of

- $\mathcal{M}^{ell}_{r,d}$; $\mathcal{M}^{ell}_{r,d,D}$ (w/. out parabolic structure)

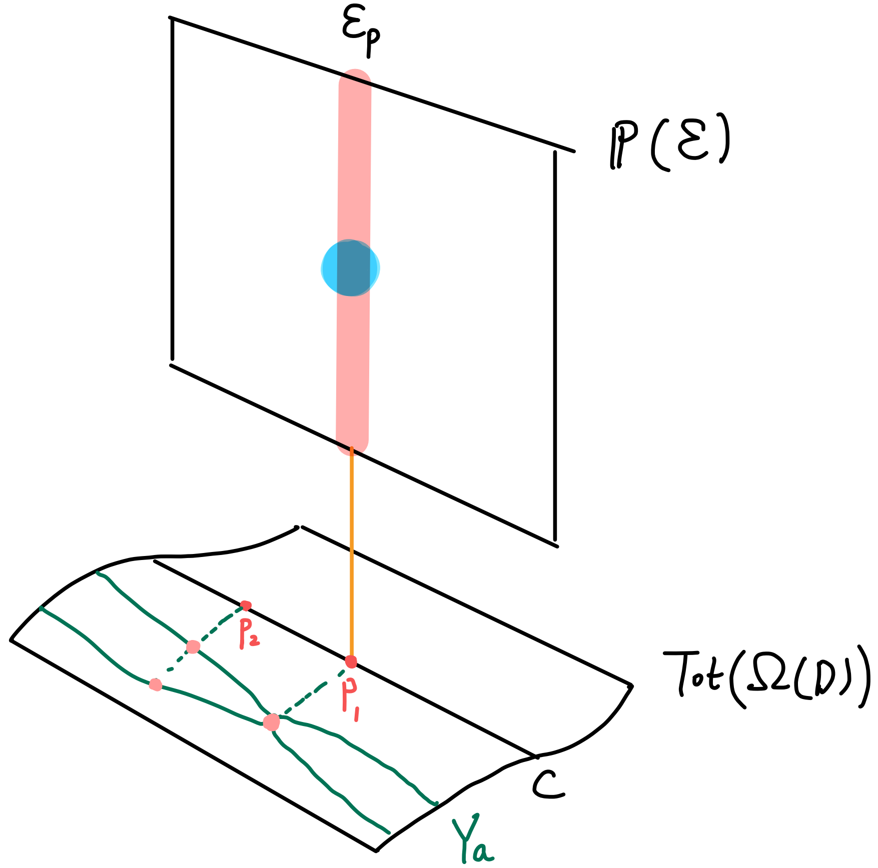
$\Rightarrow$ generators of W-alg. are in the form of $\tilde{D}_{m,n}(\pi)$, $\pi \in H^*(C) = H^*(\mathrm{Tot}_C(\Omega(D))$

$$\begin{matrix} \text{ii} & \\ H & \cong H^*_{pure}(\mathrm{Tot}_C(\Omega(D))) \end{matrix}$$

For the case of $\mathcal{M}^{parell}_{r,d,D}$. need modify $H \rightsquigarrow H_{r,D} = H^*_{orb}(C_{(r,D)})$

$$\cong H[p_i]_{p\in D,\, i=1,\dots,r} / \sim .$$

perversity $\Rightarrow \hbar$-weight



$\underset{n}{\psi_1}(P_i) := y_{P,i}^{\,n}$ comes from extra taut. class from parab. structure.

$\underset{n}{T_1}(P_i) := y_{P,i}^{\,n} X_{P,i}$ where $X_{P,i} := \mathrm{Mod}_{P,i}^*$ [Note that $\mathrm{Mod}_{P,i}$ keeps par-ell locus]

$y_{P,i}$ can also be def. for $M_{r,d,D}^{par}$ w/ i-th eigenspace ⇝ i-th subquotient in the flag.

[But Hecke operator could be destabilizing]

§ Reduction steps

Twisted parabolic case:

There is $\mathbb{C}^\times$-action on $M_{r,d}^{par}$ by scaling the Higgs field

<u>Lma</u> (Hausel - Rodriguez - Villegas) (i) $M_{r,d}^{par}$ is a semi-projective variety and thus it has pure cohomology

i.e. • $(M_{r,d,D}^{par})^{\mathbb{C}^\times}$ is proper

• $\forall x \in M_{r,d}^{par}$, $\lim_{\lambda\to 0} \lambda . x$ exists

(ii) The fibers of $M_{r,d}^{par} \xrightarrow{\chi_D} \mathcal{A}_D$ has isomorphic, pure cohomology.

(iii) As a result of (ii), $i_\mu^* : H^*(M^{par}_{r,d,D}) \longrightarrow H^*(\chi_D^{-1}(\mu))$ is an isom.

Prop: $(\iota^{parell})^* : H^*(M^{par}_{r,d,D}) \longrightarrow H^*(M^{parell}_{r,d,D})$ is injective.

Pf: Choose good enough $\mu \in A_D$, we have $\chi_D^{-1}(\mu) \subseteq M^{parell}_{r,d,D}$

So i_μ^* factors as:

$$\begin{tikzcd} H^*(M^{par}_{r,d,D}) \arrow[r] \arrow[rr, bend right, "\text{Lma above}"'] & H^*(M^{parell}_{r,d,D}) \arrow[r] & H^*(\chi_D^{-1}(\mu)) \end{tikzcd}$$

$$\sum_p Tr(res_p(\theta)) = 0$$
$$\Updownarrow$$

Concrete construct° of $\mu = (\mu_{p,i}) \in A^{r|D|}$, $\sum_{p,i} \mu_{p,i} = 0$; s.t.

(a) fixed p. $\mu_{p,i}$ distinct $i = 1, \cdots, r$

(b) no proper subset of $\{\mu_{p,i}\}$ sum to zero $\boxtimes$

As a result, we see $H^*(M^{par}_{r,d,D}) \cong H^*_{pure}(M^{parell}_{r,d,D})$

Since $M^{parell}_{r,d,D} \subseteq M^{par}_{r,d,D}$ open, the restriction map $(\iota^{parell})^*$ preserves perverse filt. also a map of Lefschetz $\Rightarrow P = C$ in this case.

i.e map of Lefschetz structure

$\boxed{\text{Classical case}}$

We first assume that $P = C$ holds for $\overset{\mathcal{A}_D}{\overset{\cup}{\chi_D^{-1}(0)}}$, i.e. $\mathrm{res}_p(\theta)$ nilpotent.

In particular, we consider $D = (p)$.

$$\Rightarrow \quad \begin{tikzcd}
\chi_D^{-1}(0) \arrow[rr, bend right, "A"'] \arrow[dr] & \tilde{M}_{r,d} \arrow[l, hook', "\iota"'] \arrow[r, "\pi"] \arrow[d] & M_{r,d} \arrow[ll, bend right, "B"'] \arrow[dl] \arrow[d] \\
& \overline{\mathcal{A}}^{par} & \mathcal{A} \arrow[l]
\end{tikzcd}$$

π: forgetful map (of parabolic structure), relative flag var.

$$\operatorname{codim}_{\chi_D^{-1}(0)}(\tilde{M}_{r,d}) = \dim \begin{pmatrix} 0 & & * \\ & \ddots & \\ & & 0 \end{pmatrix} = \binom{r}{2}$$

$$\| $$

$$\operatorname{reldim}(\pi) = \dim\Big(\underbrace{\{0 = V_0 \subsetneq V_1 \cdots \subsetneq V_p\}}_{GL_n/B}\Big) = \binom{r}{2}$$

$A = \pi_* \iota^*$

$B = \iota_* \pi^*$

Define $\Delta := \prod_{1 \le i < j \le r} (y_{p,i} - y_{p,j}) \in H^{2\binom{r}{2}}(\chi_D^{-1}(0))$

we have : • $\pi_*(\Delta) = \pm r!$ as Δ is the Euler class of the relative tangent bundle of $\tilde{M}_{r,d} \longrightarrow M_{r,d}$

• $l^* l_* = \Delta$ as it is the multiplication by Euler class of the normal bundle of $\tilde{M}_{r,d}$ in $\chi_D^{-1}(0)$

$\Rightarrow AB = \pi_* l^* l_* \pi^* \stackrel{\simeq}{=} \pm r!$

<u>Thm</u> $P=W$ holds for $M_{r,d}$. i.e. $P_m H^*(X)$ is the span of $\prod_i \psi_{m_i}(\xi)$ s.t. $\sum_i m_i \leq m+N$, $-N$ = perversity of $1 \in H^0(M_{r,d})$

<u>Pf</u>: $-N'$: perversity of $1 \in H^0(\chi_D^{-1}(0))$

perverse fil is increasing

• $A(\Delta) = \pm r! \in P_{-N'+\binom{r}{2}} H^0(M_{r,d}) \Longrightarrow -N \leq -N' + \binom{r}{2}$

• $\Delta = B(1) \in P_{-N}$

If $-N < -N' + \binom{r}{2}$ since Δ is $\hbar$-homogeneous of wt $= -N' + \binom{r}{2}$

$\Rightarrow \Delta = 0$, contradiction $\Rightarrow N' = N + \binom{r}{2}$

- First consider $f = \prod_i \psi_{m_i}(\xi_i) \in H^*(M_{r,d})$

$$Bf = \pm \Delta f \in P_{\sum m_i - N' + \binom{r}{2}} = P_{\sum_i m_i - N}$$

$$ABf = \pm r! f \in P_{\sum_i m_i - N} \Rightarrow f \in P_{\sum_i m_i - N} \qquad (\textstyle\sum m_i = m)$$

- Then for any $f \in P_m H^*(M_{r,d})$, wts it is of form of $\sum \prod_i \psi_{m_i}(\xi_i)$

$$B(f) = \Delta f = \sum_k \lambda_k g_k \in H^*_{pure}(\chi_D^{-1}(0)), \quad \text{where}$$

g_k : monomial in taut. class of form $\prod_i \psi_{m_i}(\xi_i)$. w/. $\sum_i m_i \leq N' + m$.

There is a $\mathfrak{S}_r$-permutation on $H^*_{pure}(\chi_D^{-1}(0))$ permutes $y_{p,i}$

$$\mathrm{Asym} : H^*(\chi_D^{-1}(0)) \longrightarrow H^*(\chi_D^{-1}(0)) \text{ given by } \frac{1}{r!}\sum_{\sigma \in \mathfrak{S}_r} (-1)^{|\sigma|} \cdot \sigma$$

$$\Rightarrow \quad \Delta f = \sum_k \lambda_k \boxed{\mathrm{Asym}(g_k)}$$

product of Δ w/. linear comb. of $\prod_i \psi_{m_i'}(\xi_i')$ s.t. $\xi_i \in H^*(C)$

$\sum m_i' \leq (N' - \binom{r}{2}) + m = N + m$

$\Rightarrow B(f) = \Delta f$ can be written:

$$= B(\textstyle\sum \lambda_k' g_k')$$

s.t. g_k is monomial of the form $\prod_i \psi_{m_i}(\xi_i)$. where $\sum_i m_i \leqslant N+m$

then apply A to $B(f)$ $\Rightarrow$ $f \in P_m H^*(M_{r,d})$ $\boxtimes$

In the end, we investigate

Parabolic case :

We know $H^*(\chi_D^{-1}(0)) \cong H^*(M^{par}_{r,d,D})$ — with X_0 written over $\chi_D^{-1}(0)$ and X written over $M^{par}_{r,d,D}$

consider $\bar{X} := (X \times \mathbb{C} \setminus \chi^{-1}(0) \times \{0\}) / \mathbb{C}^\times$

$U := \bar{X} \setminus \chi^{-1}(0)$
total space of line bdle over $\partial X = \bar{X} \setminus X$

$$\begin{tikzcd}
U \arrow[r, hook] & \bar{X} \arrow[ddl] & & \bar{X}_0 \arrow[ll, "\bar\iota"'] \arrow[ddl] \\
& X \arrow[u] \arrow[dd] & & X_0 \arrow[ll, "\iota"'] \arrow[u] \arrow[dd] \\
\bar{A} & & \bar{A}_0 \arrow[ll, "\bar\chi"'] & \\
& A \arrow[ul] & & A_0 \arrow[ll] \arrow[ul]
\end{tikzcd}$$

Prop : $\exists L \in H^2(\bar{X})$ hyperplane class s.t.

$$H^*(X) = H^*(\bar{X}) / LH^*(\bar{X})$$

$$[\bar{X} \setminus X] = L \in H^*(\bar{X}) \twoheadrightarrow H^*(\bar{X} \setminus X) \longrightarrow H^*(\bar{X})$$

Prop : (i) $\operatorname{Im} \bar{\iota}^* \subseteq \operatorname{Im} L^n$; $n = \dim X - \dim X_0 = N - N_0$

(ii) $\forall\, i \geq 0$, $(\iota^*)^{-1}(\operatorname{Ker} L^i + \operatorname{Im} L) \subseteq \operatorname{Ker} L^{i+n} + \operatorname{Im} L$

both $\bar{\iota}_*, \bar{\iota}^*$ commute w/ L

$\bar{\iota}_* \bar{\iota}^* = [\bar{X}_0] \in H^{2n}(\bar{X})$

$\bar{X}_0 = \bar{\chi}^{-1}(\bar{A}_0)$ $\quad \operatorname{codim}_{\bar{A}}(\bar{A}_0) = n$

$L^n = c[\bar{X}_0]$

Suppose $\alpha \in P_i H^j(X_0)$ α can be rep. in $H^*(\bar{X}_0)$ by an element in W_{i+n-j}

$W_{i+n-j} \subseteq \operatorname{Ker} L^{i+N_0-j+1} + \operatorname{Im} L \qquad N_0 = \dim X_0$

Recall weight filtration is $W_k = \sum_{i \geq \max(0,k)} \operatorname{Ker} N^{i+1} \cap \operatorname{Im} N^{i-k}$, N nilpotent operator

$\Rightarrow$ In $H^*(\bar{X})$ via $(L^*)^{-1}$; α can rep. by element of

$$\operatorname{Ker} L^{i+N_0+n-j+1} + \operatorname{Im} L = \operatorname{Ker} L^{i+N-j+1} + \operatorname{Im}$$

$\Rightarrow$ $(L^*)^{-1}\alpha \in P_i H^j(X)$.

we see that $(L^*)^{-1}$ preserves perverse filtration, so it is a map of Lefschetz structure